

SSD_DISP point screen reference

1. Basic introduction

1.1. Overview

This document mainly explains the configuration and use related to the touch screen.

1.2. Chip related specifications

Specification Chip	TTL	MIPI	SPI	SRGB	HDMI
201/202	support	support	support	not support	not support
203	not support	not support	support	not support	support



1.3. Keyword Description

- TTL/MIPI/SRGB (SSD20X does not support SRGB)

These are the LCD screen interface.

The TTL interface signal type is TTL level, the content of the signal is RGB565 or RGB666 or RGB888, horizontal and vertical synchronization (HSYNC/VSYNC), clock (CLK) and data valid signal (DE).

The MIPI interface signal type is LVDS signal, and the content of the signal is video stream data and control commands.

The SRGB interface divides the RGB data of one pixel into three transmissions to the panel.

- HDMI

High Definition Multimedia Interface (High Definition Multimedia Interface) is a fully digital video and sound transmission interface that can transmit uncompressed audio and video signals.

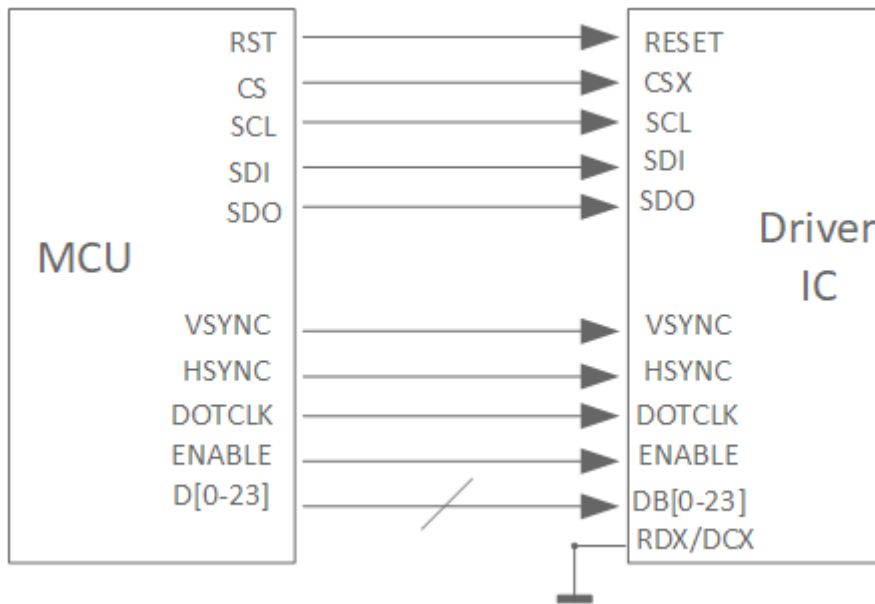
2. TTL point screen introduction

2.1. Specifications

Specification Chip	maximum resolution	Signal format
201/202	1280x720 60fps	RGB565/RGB666/RGB888
203	not support	not support

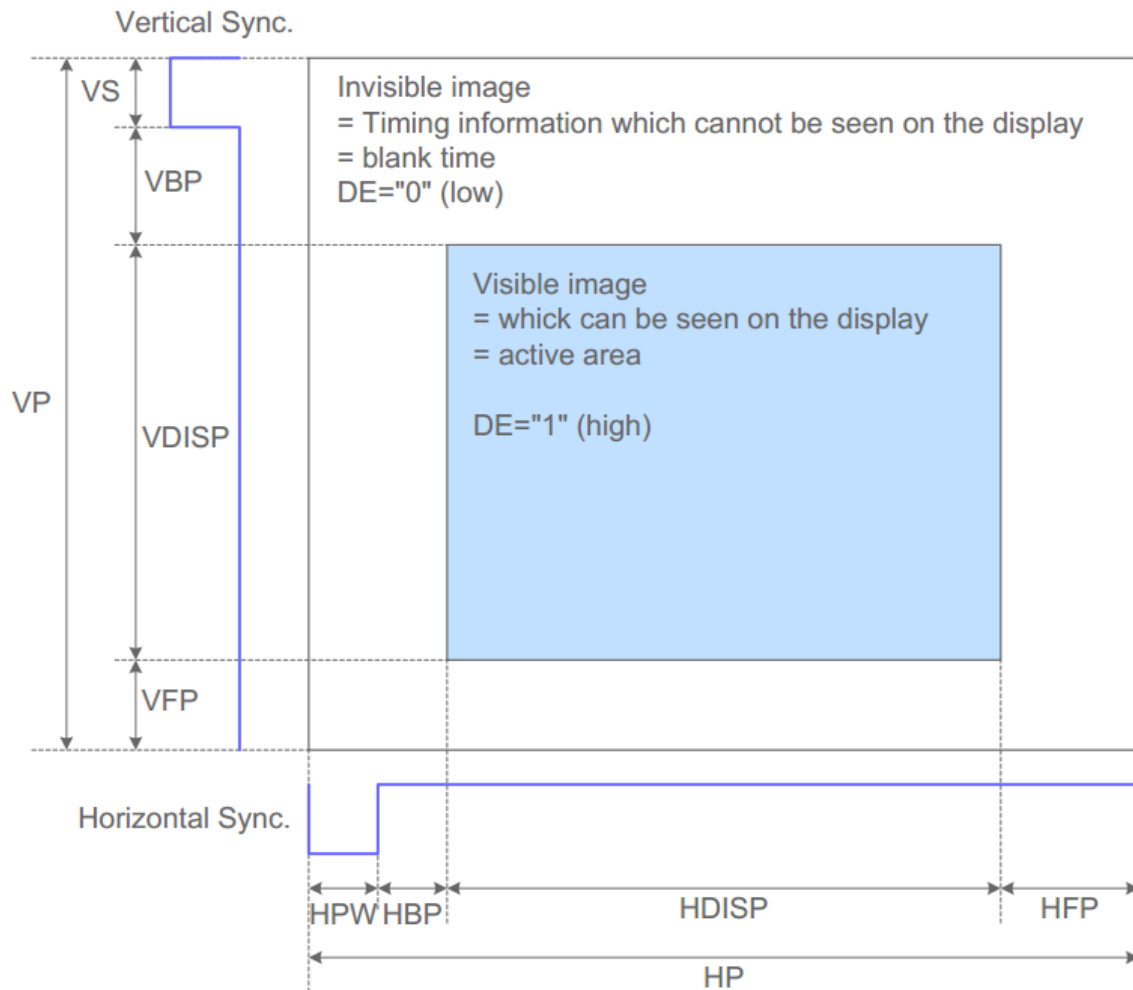
2.2. Basic Concepts

There are two types of Parallel RGB Interface, DE mode and HV mode. When DE is enabled, VSYNC, HSYNC, DOTCLK, DE, D[0-23] pins are used. When HV mode is enabled, VSYNC is used. , HSYNC, DOTCLK, D[0-23]. Now the program uses DE mode by default, please configure relevant parameters according to DE mode.



Some panel driver ICs need to be initialized, which is actually the setting of their internal register, which is usually communicated through the SPI or IIC interface. The cmd and data required for initialization are generally provided by the panel factory, and the data format when sending needs to refer to the panel datasheet.

Panels using Parallel RGB Interface require HSYNC, VSYNC, and DOTCLK as synchronization signals, and RGB data is only valid in a specific interval of timing.



The blanking interval in the line and field signals is an invisible area, and only the RGB data in the active interval will be finally displayed. Different panel Driver ICs have different requirements for the blanking interval.

In line signal:

$$H_{total} = H_{SYNC} + H_{BP} + H_{FP} + H_{Active}$$

场信号中：

$$V_{total} = V_{SYNC} + V_{BP} + V_{FP} + V_{Active}$$

最后计算像素时钟频率：

$$Pixel\ CLK = H_{total} * V_{total} * fps$$

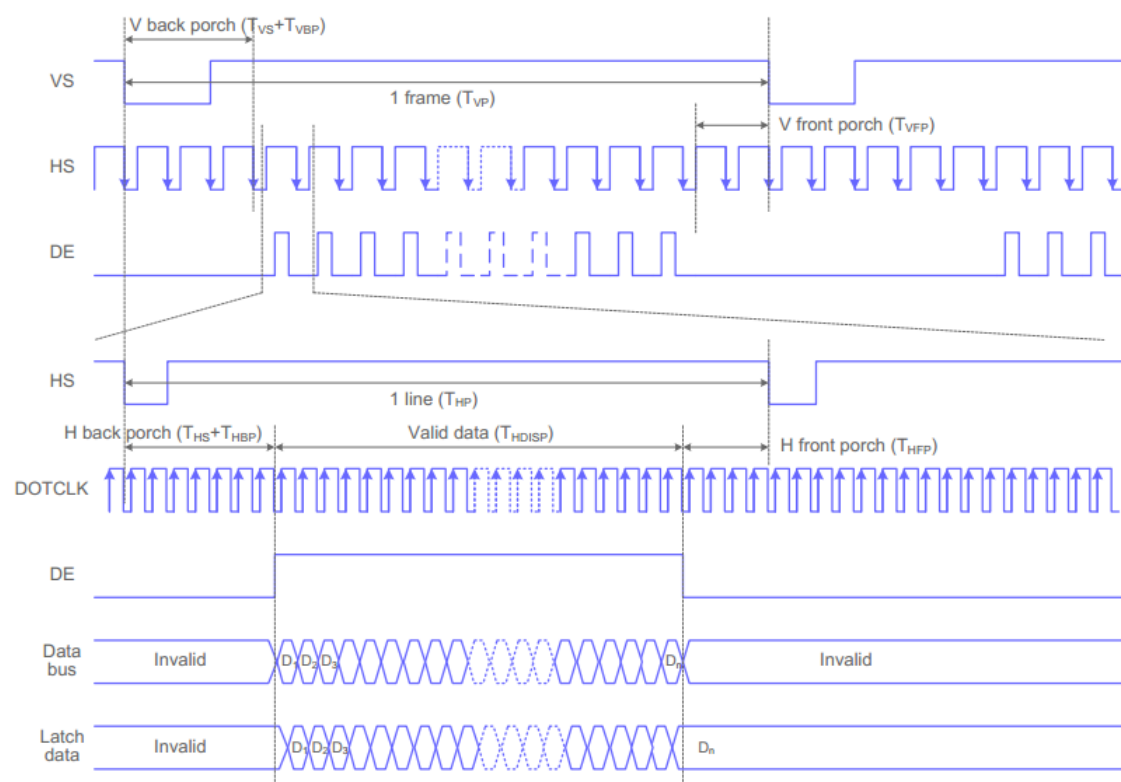
屏参中时序相关的配置主要是调整行信号中HSYNC、HBP、HFP和场信号中VSYNC、VBP、VFP。panel spec中会提供blanking区间各部分的长度要求。

blanking区间是一个可调范围，最终计算出的pixel clk也是在一个范围内。

主控芯片一般会有时钟频率的限制，所以在点一个新的panel时，应先计算panel的pixel clk，如果将要点的panel满足像素时钟频率的要求，则一般可以点起来。

Parallel RGB Interface Timing:

The timing chart of RGB interface DE mode is shown as follows.



Note: The setting of front porch and back porch in host must match that in IC as this mode.

2.3. 点屏流程



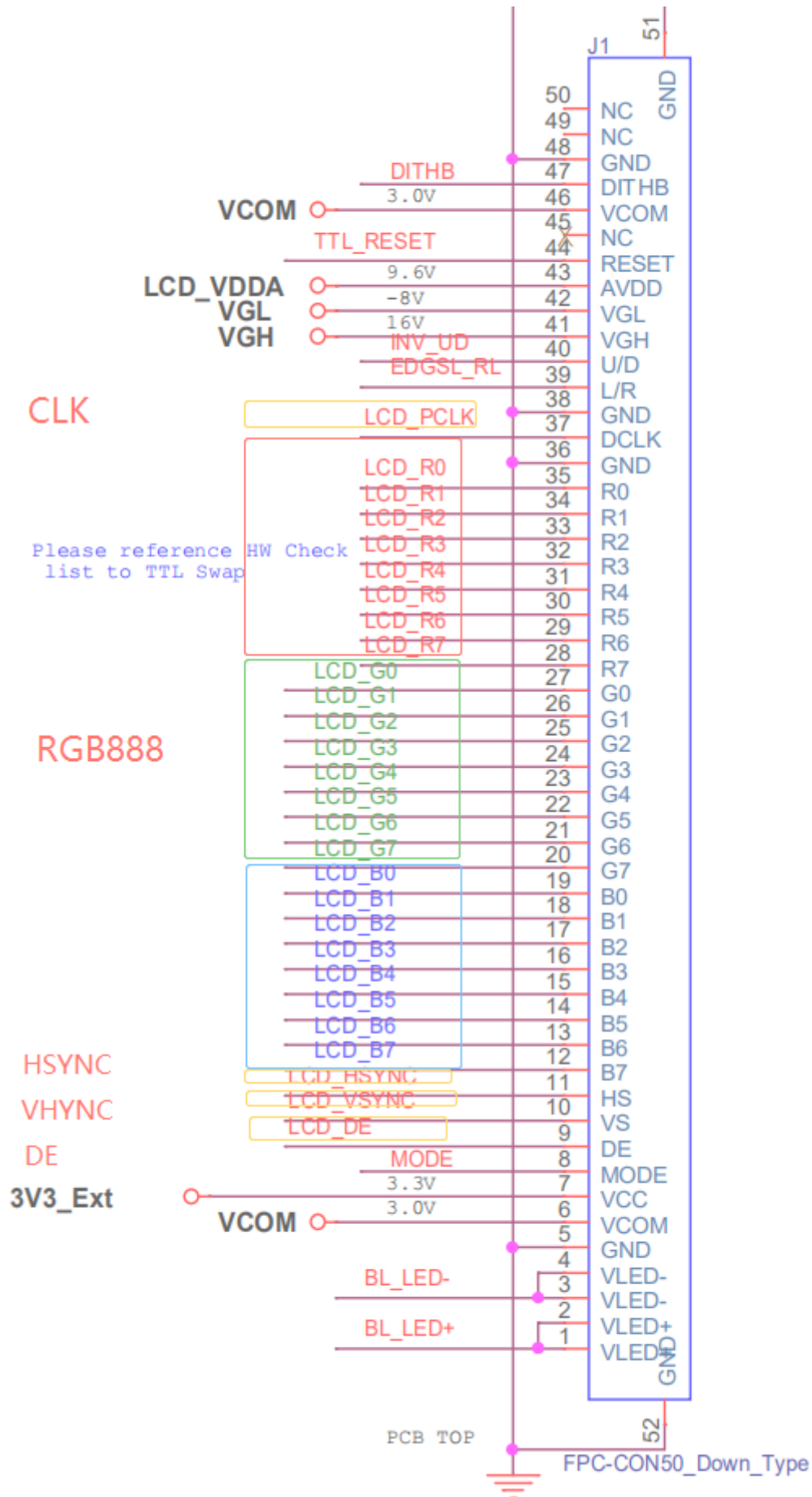
2.4. 硬件相关

2.4.1. 查看原理图

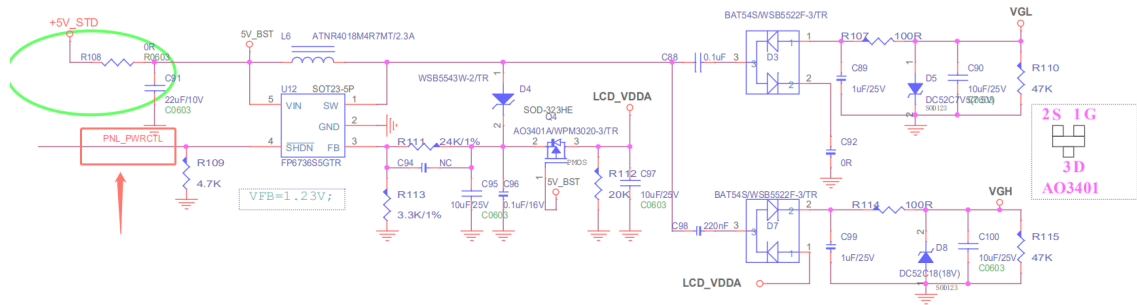
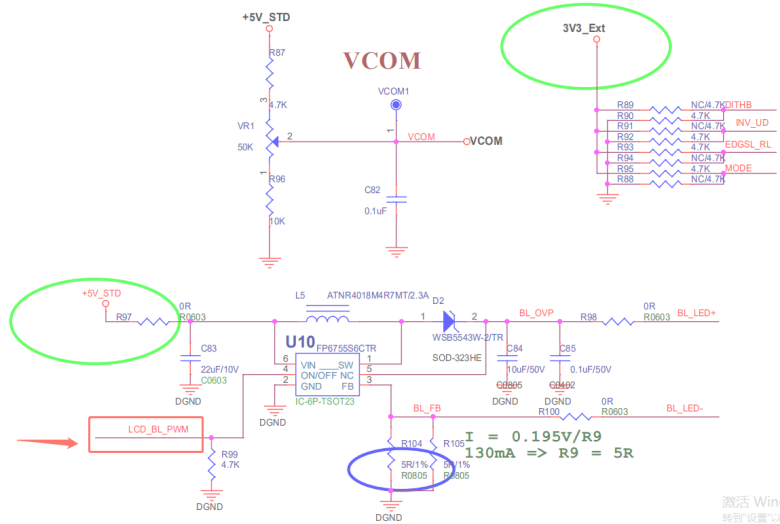
以SSD202 demo board原理图举例说明。

查看如下原理图可知RGB888数据格式，以及CLK/HSYNC/VSYNC/DE对应的Pin：

F



TP_SQ_L	≡≡	TP_SQ_L		
TP_SQ_A	≡≡	TP_SQ_A		
TP_INT	≡≡	TP_INT		
TP_RESET	≡≡	TP_RESET		
PNE_PWRCSTL	≡≡	PNE_PWRCSTL		
LCD_BL_PWM	≡≡	LCD_BL_PWM		
	≡≡	LCD_R0	≡≡	LCD_R0 4
	≡≡	LCD_R1	≡≡	LCD_R1 4
	≡≡	LCD_R2	≡≡	LCD_R2 4
	≡≡	LCD_R3	≡≡	LCD_R3 4
	≡≡	LCD_R4	≡≡	LCD_R4 4
	≡≡	LCD_R5	≡≡	LCD_R5 4
	≡≡	LCD_R6	≡≡	LCD_R6 4
	≡≡	LCD_R7	≡≡	LCD_R7 4
	≡≡	LCD_G0	≡≡	LCD_G0 4
	≡≡	LCD_G1	≡≡	LCD_G1 4
	≡≡	LCD_G2	≡≡	LCD_G2 4
	≡≡	LCD_G3	≡≡	LCD_G3 4
	≡≡	LCD_G4	≡≡	LCD_G4 4
	≡≡	LCD_G5	≡≡	LCD_G5 4
	≡≡	LCD_G6	≡≡	LCD_G6 4
	≡≡	LCD_G7	≡≡	LCD_G7 4
	≡≡	LCD_B0	≡≡	LCD_B0 4
	≡≡	LCD_B1	≡≡	LCD_B1 4
	≡≡	LCD_B2	≡≡	LCD_B2 4
	≡≡	LCD_B3	≡≡	LCD_B3 4
	≡≡	LCD_B4	≡≡	LCD_B4 4
	≡≡	LCD_B5	≡≡	LCD_B5 4
	≡≡	LCD_B6	≡≡	LCD_B6 4
	≡≡	LCD_B7	≡≡	LCD_B7 4
	≡≡	LCD_DE	≡≡	LCD_DE 4
	≡≡	LCD_VSYNC	≡≡	LCD_VSYNC 4
	≡≡	LCD_HSYNC	≡≡	LCD_HSYNC 4
	≡≡	LCD_PCLK	≡≡	LCD_PCLK 4
	≡≡	TTL_RESET	≡≡	TTL_RESET 4
	≡≡	PSPIH_CS	≡≡	PSPIH_CS 4
	≡≡	PSPIH_CLK	≡≡	PSPIH_CLK 4
	≡≡	PSPIH_MOSI[0]	≡≡	PSPIH_MOSI[0] 4



芯片端对应pin脚如下：

Series Resistance TTL ESD Protect

PAD_TTL0	R171	100R	LCD_R0	LCD_R0	9
PAD_TTL1	R169	100R	LCD_R1	LCD_R1	9
PAD_TTL2	R173	100R	LCD_R2	LCD_R2	9
PAD_TTL3	R170	100R	LCD_R3	LCD_R3	9
PAD_TTL4	R204	100R	LCD_R4	LCD_R4	9
PAD_TTL5	R202	100R	LCD_R5	LCD_R5	9
PAD_TTL6	R175	100R	LCD_R6	LCD_R6	9
PAD_TTL7	R172	100R	LCD_R7	LCD_R7	9
PAD_TTL8	R176	100R	LCD_G0	LCD_G0	9
PAD_TTL9	R174	100R	LCD_G1	LCD_G1	9
PAD_TTL10	R179	100R	LCD_G2	LCD_G2	9
PAD_TTL11	R177	100R	LCD_G3	LCD_G3	9
PAD_TTL12	R181	100R	LCD_G4	LCD_G4	9
PAD_TTL13	R178	100R	LCD_G5	LCD_G5	9
PAD_TTL14	R182	100R	LCD_G6	LCD_G6	9
PAD_TTL15	R180	100R	LCD_G7	LCD_G7	9
PAD_TTL16	R186	100R	LCD_B0	LCD_B0	9
PAD_TTL17	R196	100R	LCD_B1	LCD_B1	9
PAD_TTL18	R199	100R	LCD_B2	LCD_B2	9
PAD_TTL19	R197	100R	LCD_B3	LCD_B3	9
PAD_TTL20	R201	100R	LCD_B4	LCD_B4	9
PAD_TTL21	R198	100R	LCD_B5	LCD_B5	9
PAD_TTL22	R203	100R	LCD_B6	LCD_B6	9
PAD_TTL23	R200	100R	LCD_B7	LCD_B7	9
PAD_TTL24	R150	100R	LCD_PCLK	LCD_PCLK	9
PAD_TTL25	R151	100R	LCD_HSYNC	LCD_HSYNC	9
PAD_TTL26	R152	100R	LCD_VSYNC	LCD_VSYNC	9
PAD_TTL27	R153	100R	LCD_DE	LCD_DE	9

2.4.2. 对应Padmux

查看SSD202 HW CheckList找到管脚对应的padmux，如下：

SSD201 HW Checklist.xlsx - Excel

页面布局 公式 数据 审阅 视图 福昕PDF

xinhua.ye(叶信华)

	C	D	L	M	N	O	P	Q	R	S
	Ball Pin Name	GPIO index	reg_eth1_mode	reg_eth1_mode	reg_ttl_mode	reg_ttl_mode	reg_ttl_mode	reg_ttl_mode	reg_ttl_mode	reg_sdio_mode
			3	5	1	7	10	12	13	1
		9pin	9pin		28pin	28pin	20pin	28pin	20pin	6pin
	PAD_UART1_TX	50						TTL_DE	TTL_DE	
	PAD_FUART_RX	15						TTL_VSYNC	TTL_VSYNC	
	PAD_FUART_TX	16						TTL_HSYNC	TTL_HSYNC	
	PAD_FUART_CT	17						TTL_CK	TTL_CK	
	PAD_FUART_RT	18						TTL_DOUT[0]	TTL_DOUT[0]	
	PAD_TTL0	19			TTL_DOUT[0]	TTL_DE	TTL_DE	TTL_DOUT[1]	TTL_DOUT[1]	
	PAD_TTL1	20			TTL_DOUT[1]	TTL_VSYNC	TTL_VSYNC	TTL_DOUT[2]	TTL_DOUT[2]	
	PAD_TTL2	21			TTL_DOUT[2]	TTL_HSYNC	TTL_HSYNC	TTL_DOUT[3]	TTL_DOUT[3]	
	PAD_TTL3	22			TTL_DOUT[3]	TTL_CK	TTL_CK			
	PAD_TTL4	23			TTL_DOUT[4]	TTL_DOUT[0]	TTL_DOUT[0]			
	PAD_TTL5	24			TTL_DOUT[5]	TTL_DOUT[1]	TTL_DOUT[1]			
	PAD_TTL6	25			TTL_DOUT[6]	TTL_DOUT[2]	TTL_DOUT[2]	TTL_DOUT[4]	TTL_DOUT[4]	
	PAD_TTL7	26			TTL_DOUT[7]	TTL_DOUT[3]	TTL_DOUT[3]	TTL_DOUT[5]	TTL_DOUT[5]	
	PAD_TTL8	27			TTL_DOUT[8]	TTL_DOUT[4]	TTL_DOUT[4]	TTL_DOUT[6]	TTL_DOUT[6]	
	PAD_TTL9	28			TTL_DOUT[9]	TTL_DOUT[5]	TTL_DOUT[5]	TTL_DOUT[7]	TTL_DOUT[7]	
	PAD_TTL10	29			TTL_DOUT[10]	TTL_DOUT[6]	TTL_DOUT[6]	TTL_DOUT[8]	TTL_DOUT[8]	
	PAD_TTL11	30			TTL_DOUT[11]	TTL_DOUT[7]	TTL_DOUT[7]	TTL_DOUT[9]	TTL_DOUT[9]	
	PAD_TTL12	31			TTL_DOUT[12]	TTL_DOUT[8]	TTL_DOUT[8]	TTL_DOUT[10]	TTL_DOUT[10]	
	PAD_TTL13	32			TTL_DOUT[13]	TTL_DOUT[9]	TTL_DOUT[9]	TTL_DOUT[11]	TTL_DOUT[11]	
	PAD_TTL14	33			TTL_DOUT[14]	TTL_DOUT[10]	TTL_DOUT[10]	TTL_DOUT[12]	TTL_DOUT[12]	
	PAD_TTL15	34			TTL_DOUT[15]	TTL_DOUT[11]	TTL_DOUT[11]	TTL_DOUT[13]	TTL_DOUT[13]	
	PAD_TTL16	35	ETH1_MDIO		TTL_DOUT[16]	TTL_DOUT[12]	TTL_DOUT[12]	TTL_DOUT[14]	TTL_DOUT[14]	
	PAD_TTL17	36	ETH1_MDC		TTL_DOUT[17]	TTL_DOUT[13]	TTL_DOUT[13]	TTL_DOUT[15]	TTL_DOUT[15]	
	PAD_TTL18	37	ETH1_COL		TTL_DOUT[18]	TTL_DOUT[14]	TTL_DOUT[14]	TTL_DOUT[16]		
	PAD_TTL19	38	ETH1_RXD0		TTL_DOUT[19]	TTL_DOUT[15]	TTL_DOUT[15]	TTL_DOUT[17]		
	PAD_TTL20	39	ETH1_RXD1		TTL_DOUT[20]	TTL_DOUT[16]		TTL_DOUT[18]		
	PAD_TTL21	40	ETH1_TX_CLK	ETH1_TXD1	TTL_DOUT[21]	TTL_DOUT[17]		TTL_DOUT[19]		
	PAD_TTL22	41	ETH1_TXD0	ETH1_TXD0	TTL_DOUT[22]	TTL_DOUT[18]		TTL_DOUT[20]		
	PAD_TTL23	42	ETH1_TXD1	ETH1_TX_EN	TTL_DOUT[23]	TTL_DOUT[19]		TTL_DOUT[21]		
	PAD_TTL24	43	ETH1_TX_EN	ETH1_TX_CLK	TTL_CK	TTL_DOUT[20]		TTL_DOUT[22]		
	PAD_TTL25	44		ETH1_COL	TTL_HSYNC	TTL_DOUT[21]		TTL_DOUT[23]		
	PAD_TTL26	45		ETH1_RXD0	TTL_VSYNC	TTL_DOUT[22]				
	PAD_TTL27	46		ETH1_RXD1	TTL_DE	TTL_DOUT[23]				

2.5. 软件相关

2.5.1. 配置padmux

通过如上可知panel使用的是ttl28_mode_1

1. user space配置找到项目对应使用的*padmux.dts配置如下：

```
ty2m-ssc011a-s01a-padmux-display.dtsi - Notepad++
ge Settings Tools Macro Run Plugins Window ?
infinity2m-ssc011a-s01a-padmux-display.dtsi
43 //<PAD_FUART_TX PINMUX_FOR_GPIO_MODE MDRV_PUSE_UTMI_POWER>,
44 <PAD_FUART_CTS PINMUX_FOR_GPIO_MODE MDRV_PUSE_CPUFREQ_VID0>,
45 <PAD_FUART_RTS PINMUX_FOR_GPIO_MODE MDRV_PUSE_CPUFREQ_VID1>,
46 <PAD_TTL0 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT00 >,
47 <PAD_TTL1 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT01 >,
48 <PAD_TTL2 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT02 >,
49 <PAD_TTL3 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT03 >,
50 <PAD_TTL4 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT04 >,
51 <PAD_TTL5 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT05 >,
52 <PAD_TTL6 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT06 >,
53 <PAD_TTL7 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT07 >,
54 <PAD_TTL8 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT08 >,
55 <PAD_TTL9 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT09 >,
56 <PAD_TTL10 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT10 >,
57 <PAD_TTL11 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT11 >,
58 <PAD_TTL12 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT12 >,
59 <PAD_TTL13 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT13 >,
60 <PAD_TTL14 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT14 >,
61 <PAD_TTL15 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT15 >,
62 <PAD_TTL16 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT16 >,
63 <PAD_TTL17 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT17 >,
64 <PAD_TTL18 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT18 >,
65 <PAD_TTL19 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT19 >,
66 <PAD_TTL20 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT20 >,
67 <PAD_TTL21 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT21 >,
68 <PAD_TTL22 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT22 >,
69 <PAD_TTL23 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DOUT23 >,
70 <PAD_TTL24 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_CLK >,
71 <PAD_TTL25 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_HSYNC >,
72 <PAD_TTL26 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_VSYNC >,
73 <PAD_TTL27 PINMUX_FOR_TTL_MODE_1 MDRV_PUSE_TTL_DE >,
```

2. Uboot下padmux配置

TTL支持的RGB888/RGB666/RGB565在Uboot下已经根据屏参的Lane num自动切到对应Mode，用户可以不用关注，直接使用即可(如pin Mode不一致请找对应FAE咨询修改)

```
Uboot Project - Source Insight 4.0 - [hal_pnl.c (Y:\alkaid_mp\boot\drivers\mstar\panel\hal\infinity2m\src)]
File Edit Search Project Options Tools View Window Help
Search Project: Search Project (Alt+Shift+P)
Advanced... Match Case Word Variations Word Fragments
hal_pnl.c (Y:\alkaid_mp\boot\drivers\mstar\panel\hal\infinity2m\src) x hal_pnl_if.c (Y:\alkaid_mp\boot\drivers\mstar\panel\hal\infinity2m\src) hal_pnl_st.h (Y:\alkaid_mp\boo
hal_pnl.c
mbol Name (Alt+L)
ul6LoopGainDsi
ul6LoopDivVdsi
ul6LoopGain
ul6LoopDiv
HalPnlGetMipiUi
HalPnlGetLp1Idx
HalPnlGetLp1Gain
HalPnlGetLp1Div
HalPnlSetTTLPadMux
HalPnlSetTclPadCotnrol
HalPnlSetMipiDisPadMux
HalPnlDumpLp1Setting
HalPnlSetLp1Set
HalPnlSetVSyncSt
HalPnlSetVSyncEnd
HalPnlSetVfdeSt
HalPnlSetVfdeEnd
HalPnlSetVdeSt
HalPnlSetVdeEnd
293:
294: void HalPnlSetTTLPadMux(HalPnlOutputFormatBitMode_e enFmt)
295: {
296:     if(enFmt == E_HAL_PNL_OUTPUT_565BIT_MODE)
297:     {
298:         W2BYTESMSK(REG_CHIPTOP_OF_L,0x0000,0xFFFF);
299:         W2BYTESMSK(REG_CHIPTOP_OE_L,0x0000,0xFFFF);
300:         W2BYTESMSK(REG_CHIPTOP_OD_L,0x0D00,0xFFFF); // TTL Mode = 13
301:     }
302:     else if(enFmt == E_HAL_PNL_OUTPUT_8BIT_MODE)
303:     {
304:         W2BYTESMSK(REG_CHIPTOP_OF_L,0x0000,0xFFFF);
305:         W2BYTESMSK(REG_CHIPTOP_OE_L,0x0000,0xFFFF);
306:         W2BYTESMSK(REG_CHIPTOP_OD_L,0x0100,0xFFFF); // TTL Mode = 1
307:     }
308:     else if(enFmt == E_HAL_PNL_OUTPUT_6BIT_MODE)
309:     {
310:         W2BYTESMSK(REG_CHIPTOP_OF_L,0x0000,0xFFFF);
311:         W2BYTESMSK(REG_CHIPTOP_OE_L,0x0000,0xFFFF);
312:         W2BYTESMSK(REG_CHIPTOP_OD_L,0x0400,0xFFFF); // TTL Mode = 4
313:     }
314: } // end HalPnlSetTTLPadMux
315:
```

2.5.2. 配置屏参

- 新增屏参

- a. SSD20X采用头文件的形式配置屏参，先找一个公版Demo带的屏参文件，这里拿公版TTL的屏参SAT070CP50_1024x600.h举例(该头文件易见于公版各显示Demo)
- b. 确认屏规格书，找到对应Timing规格，如：

6.2 Timing Conditions

Item	Symbol	Values			Unit	Remark
		Min	Typ	Max		
Horizontal Display Area	thd	-	1024	-	DCLK	
DCLK frequency	fck	44.9	51.2	63	MHz	
One horizontal line	th	1200	1344	1400	DCLK	
HS pulse width	thpw	1	-	140	DCLK	
HS Blanking	thb	160	160	160	DCLK	
HS Front Porch	thfp	16	160	216	DCLK	

Item	Symbol	Values			Unit	Remark
		Min	Typ	Max		
Vertical display Area	tv	-	600	-	TH	
VS period time	tv	624	635	750	TH	
VS pulse width	tvpw	1	-	20	TH	
VS Blanking	tvb	23	23	23	TH	
VS Front Porch	tvfp	1	12	127	TH	

$$HTotal = u16Width + u16HSyncWidth + u16HSyncBackPorch + HSyncFrontPorch$$
$$VTotal = u16Heigth + u16VSyncWidth + u16VSyncBackPorch + HVyncFrontPorch$$
$$DCLK = HTotal * VTotal * FPS$$

- c. 参考以下屏参参数说明，将屏规格对应的参数填写到头文件中，然后点屏include对应头文件即可

参数	描述
m_pPanelName	panel name
m_bPanelDither	1:enable Dither 0:disable Dither
m_ePanelLinkType	接口类型 TTL :0 MIPI_DSI :10
m_bPanelInvDCLK	Pixel clk极性反转
m_bPanelInvDE	DE极性反转
m_bPanelInvHSync	HSync极性反转

m_bPanelInvVSync	Vsync极性反转
m_wPanelHSyncWidth	行同步信号脉宽
m_wPanelHSyncBackPorch	行同步信号后肩
m_wPanelVSyncWidth	场同步信号脉宽
m_wPanelVBackPorch	场同步信号后肩
m_wPanelHStart	m_wPanelHSyncWidth+m_wPanelHSyncBackPorch
m_wPanelVStart	m_wPanelVSyncWidth+ m_wPanelVBackPorch
m_wPanelWidth	行有效像素点数
m_wPanelHeight	场有效行数
m_wPanelHTotal	m_wPanelWidth+m_wPanelHSyncWidth+m_wPanelHSyncBackPorch
m_wPanelVTotal	m_wPanelHeight+m_wPanelVSyncWidth+ m_wPanelVBackPorch
m_wPanelDCLK	m_wPanelHTotal*m_wPanelVTotal*fps (IP内部使用clk)
m_wSpreadSpectrumFreq	时钟延展幅度调制 (详见附件： 展频计算v2.xlsx [media/展频计算v2.xlsx])
m_wSpreadSpectrumRatio	时钟延展频率调制 (详见附件： 展频计算v2.xlsx [media/展频计算v2.xlsx])
m_eOutputFormatBitMode	10BIT :0 //No use 6BIT :1 //RGB666 8BIT :2 //RGB888 565 :3
m_ucPanelSwapChnR	Swap Channel R 0:default 1:select R 2:select G 3:select B
m_ucPanelSwapChnG	Swap Channel G 0:default 1:select R 2:select G 3:select B
m_ucPanelSwapChnB	Swap Channel B 0:default 1:select R 2:select G 3:select B
m_ucPanelSwapRgbML	Swap Rgb MSB/LSB 0:disable M/L swap 1:enable M/L swap

如果是mipi panel还需要对mipi dsi进行配置(对应第3章MIPI点屏相关) ,
MIPI DSI参数说明如下：

参数	描述
m_wHsTrail	Default: 5
m_wHsPrpr	Default: 3
m_wHsZero	Default: 5
m_wClkHsPrpr	Default: 10
m_wClkHsExit	Default: 14
m_wClkTrail	Default: 3
m_wClkZero	Default: 12
m_wClkHsPost	Default: 10
m_wDaHsExit	Default: 5
m_wContDet	Default:0
m_wLpx	Default:16
m_wTaGet	Default:26
m_wTaSure	Default:24
m_wTaGo	Default:50
m_wHactive	Follow 屏参设定
m_wHpw	Follow 屏参设定
m_wHbp	Follow 屏参设定
m_wHfp	Follow 屏参设定
参数	描述

m_wVactive	Follow 屏参设定
m_wVpw	Follow 屏参设定
m_wVbp	Follow 屏参设定
m_wVfp	Follow 屏参设定
m_wBllp	0
m_wFps	Default:60
m_eLaneNum	one lane :1 two lane :2 three lane :3 four lane :4
m_eFormat	RGB565 :0 RGB666 :1 LOOSELY_RGB666 :2 RGB888 :3
m_eCtrlMode	CMD_MODE :0 SYNC_PULSE :1 SYNC_EVENT :2 BURST_MODE :3
m_ucClkLane	Clk lane selection(default:2) 0:select chn0 1:select chn1 2:select chn2 3:select chn3 4:select chn4
m_ucDataLane0	data lane0 selection(default:4) 0:select chn0 1:select chn1 2:select chn2 3:select chn3 4:select chn4
m_ucDataLane1	data lane1 selection(default:3) 0:select chn0 1:select chn1 2:select chn2 3:select chn3 4:select chn4
m_ucDataLane2	data lane2 selection(default:1) 0:select chn0 1:select chn1 2:select chn2 3:select chn3 4:select chn4
m_ucDataLane3	data lane3 selection(default:0) 0:select chn0 1:select chn1 2:select chn2 3:select chn3 4:select chn4
m_pCmdBuff	Mipi panel cmd m_pCmdBuff= { Cmd, parameter cnt, parameter0, parameter1, ... , Cmd, parameter cnt, parameter0, parameter1, ... , Cmd, parameter cnt, parameter0, parameter1, ... , }
m_ucPolCh0	Chn0极性 0 : default 1 : positive 2 : negative
m_ucPolCh1	Chn1极性 0 : default 1 : positive 2 : negative

m_ucPolCh2	Chn2极性 0 : default 1 : positive 2 : negative
m_ucPolCh3	Chn3极性 0 : default 1 : positive 2 : negative
m_ucPolCh4	Chn4极性 0 : default 1 : positive 2 : negative

- 代码示例

```

1. #include "SAT070CP50_1024x600.h"
2.
3. int sstar_disp_init(MI_DISP_PubAttr_t *pstDispPubAttr, int
disp_port_num)
4. {
5.     MI_PANEL_LinkType_e eLinkType;
6.
7.     if (pstDispPubAttr->eIntfType == E_MI_DISP_INTF_LCD)
8.     {
9.         pstDispPubAttr->stSyncInfo.u16Vact =
stPanelParam.u16Height;
10.        pstDispPubAttr->stSyncInfo.u16Vbb =
stPanelParam.u16VSyncBackPorch;
11.        pstDispPubAttr->stSyncInfo.u16Vfb =
stPanelParam.u16VTotal - (stPanelParam.u16VSyncWidth +
12.        stPanelParam.u16Height + stPanelParam.u16VSyncBackPorch);
13.        pstDispPubAttr->stSyncInfo.u16Hact =
stPanelParam.u16Width;
14.        pstDispPubAttr->stSyncInfo.u16Hbb =
stPanelParam.u16HSyncBackPorch;
15.        pstDispPubAttr->stSyncInfo.u16Hfb =
stPanelParam.u16HTotal - (stPanelParam.u16HSyncWidth +
16.        stPanelParam.u16Width + stPanelParam.u16HSyncBackPorch);
17.        pstDispPubAttr->stSyncInfo.u16Bvact = 0;
18.        pstDispPubAttr->stSyncInfo.u16Bvbb = 0;
19.        pstDispPubAttr->stSyncInfo.u16Bvfb = 0;
20.        pstDispPubAttr->stSyncInfo.u16Hpw =
stPanelParam.u16HSyncWidth;
21.        pstDispPubAttr->stSyncInfo.u16Vpw =
stPanelParam.u16VSyncWidth;
22.        pstDispPubAttr->stSyncInfo.u32FrameRate =
stPanelParam.u16DCLK * 1000000 / (stPanelParam.u16HTotal *
stPanelParam.u16VTotal);
23.        pstDispPubAttr->eIntfSync = E_MI_DISP_OUTPUT_USER;
24.        pstDispPubAttr->eIntfType = E_MI_DISP_INTF_LCD;

```

```
25.         eLinkType = stPanelParam.eLinkType;
26.
27.         MI_DISP_SetPubAttr(0, pstDispPubAttr);
28.         MI_DISP_Enable(0);
29.
30.         MI_PANEL_Init(eLinkType);
31.         MI_PANEL_SetPanelParam(&stPanelParam);
32.     }
33.     return 0;
34. }
```

2.6. Debug方法

cat /proc/mi_modules/mi_panel/mi_panel0

```
/ # cat /proc/mi_modules/mi_panel/mi_panel0
PanelName:SAT070CP50_800x480_60

----- PANEL Dev0 Info -----
LVDS_POL      LVDS_CH      LINK_TYPE      TI_MODE
    0           0           TTL             1
SW_ODD        SW_EVEN      SW_ODD_RB      SW_EVEN_RB
    0           0           0              0
H_Total       V_Total      Width          Height        H_Start      V_Start
1344          635         1024          600           98           27
 hbp          hspw         hfp           vbp           vspw         vfp
  46           48         226           23            4            8
DClk          FrameRate    INV_DCLK       INV_DE        InvHSync     InvVSync
  51           0           0              0             0            0
SSC_Enable    SSC_Step      SSC_Span       TI_BIT        Format        chnswap
enable        25          192            8BIT          8BIT         (0,1,2,3,4)
```

3. MIPI点屏介绍

3.1. 规格

规格 芯片	MIPI
201/201	支持
203	不支持

MIPI DSI 规格：

- 1-4 data lanes, 1 clock lane
- Level
LP: 0~12V
HS: 100~300mV
- HS: 80Mbps ~ 1.5Gbps/lane
- Pixel format
16 bpp (5,6,5 RGB) each pixel using two bytes
18 bpp (6,6,6 RGB) packed
18 bpp (6, 6, 6 RGB) loosely packed into three bytes
24 bpp (8, 8, 8 RGB), each pixel using three bytes
- video mode: BURST_MODE/SYNC_EVENT/SYNC_PULSE
- data/clock channel swap
- data/clock channel P/N swap
- data clock skew adjustment

3.2. 基本概念

MIPI DPHY timing chart:

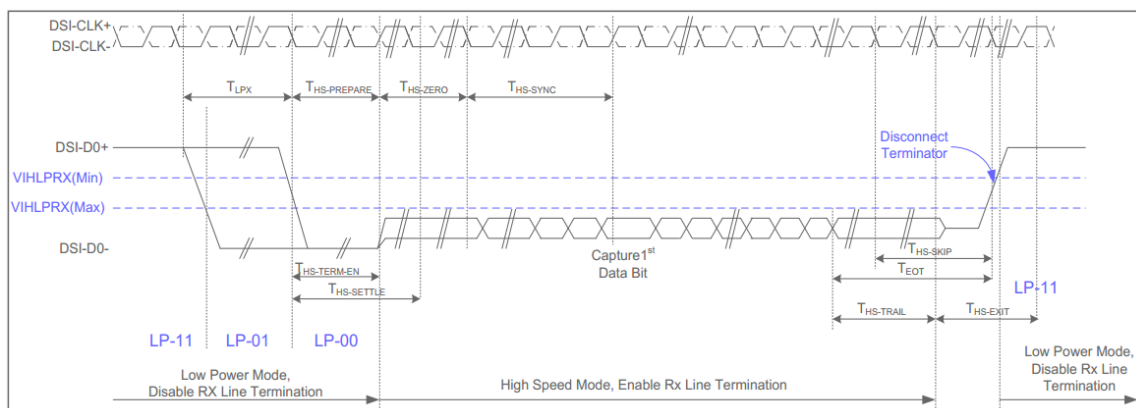


图3-1 Data lanes-Low Power Mode to/from High Speed Mode Timing

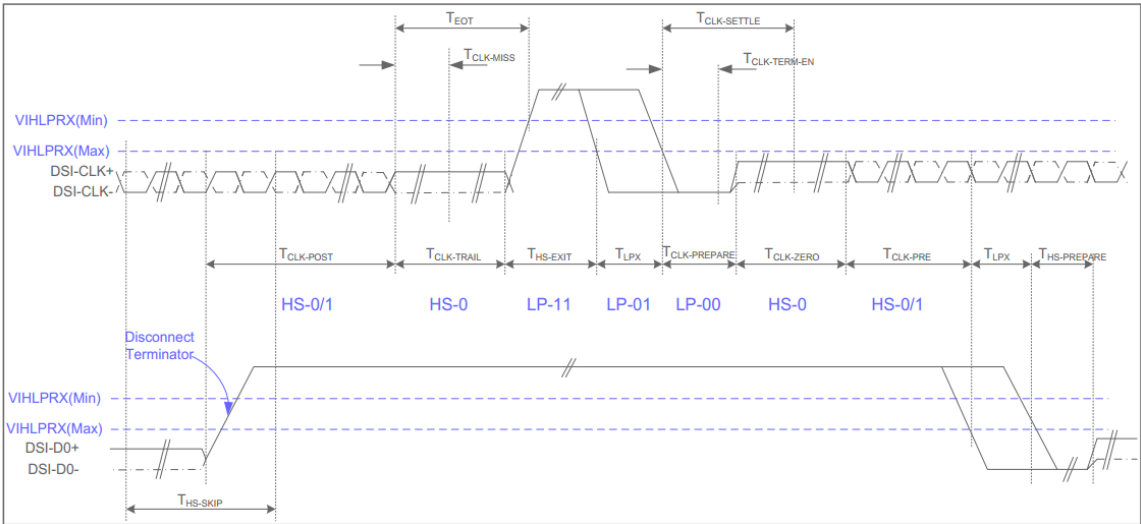


图3-2 Clock lanes- High Speed Mode to/from Low Power Mode Timing

- Configure HS Timing Parameter

HS_TRAIL/HS_EXIT

/HS_PRPR/HS_ZERO/CLK_PRPR/CLK_ZERO/CLK_POST/CLK_TRAIL

	Timing specification
THS-PREPARE + THS-zero	145ns+10*UI
THS-PREPARE	40ns+4*UI ~ 85ns +6*UI
THS-ZERO	>60ns+4*UI ~105ns+6*UI

UI表示时间间隔，等于时钟通道上任何HS状态的持续时间。

UI的计算方法：

$$H_Total = HACT + HPW + HBP + HFP$$

$$V_Total = VACT + VPW + VBP + VFP$$

$$BitsPerPixel = 24(\text{RGB888}) / 18(\text{RGB666}) / 16(\text{RGB565})$$

$$Bitrate = (H_Total) * (V_Total) * FPS * BitsPerPixel / lane \text{ number}$$

$$UI = 1 / Bitrate$$

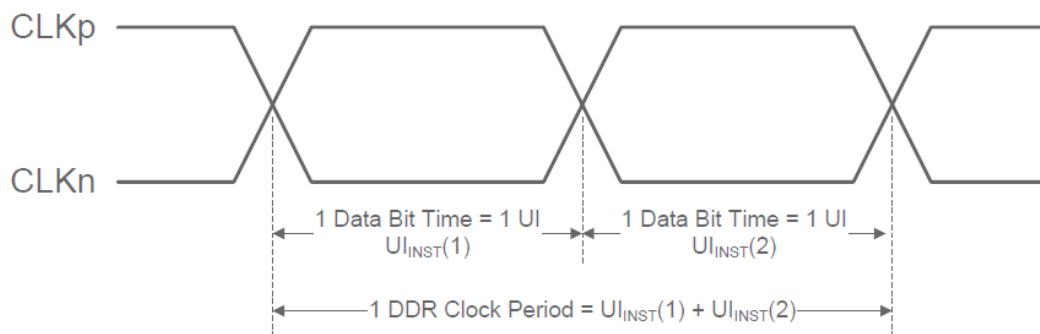


Figure 52 DDR Clock Definition

图3-3 DSI clock channel timing

如果Bitrate = 750Mps , UI = 1/Bitrate = 1.333ns , 那么HS timing parameter 计算结果为 :

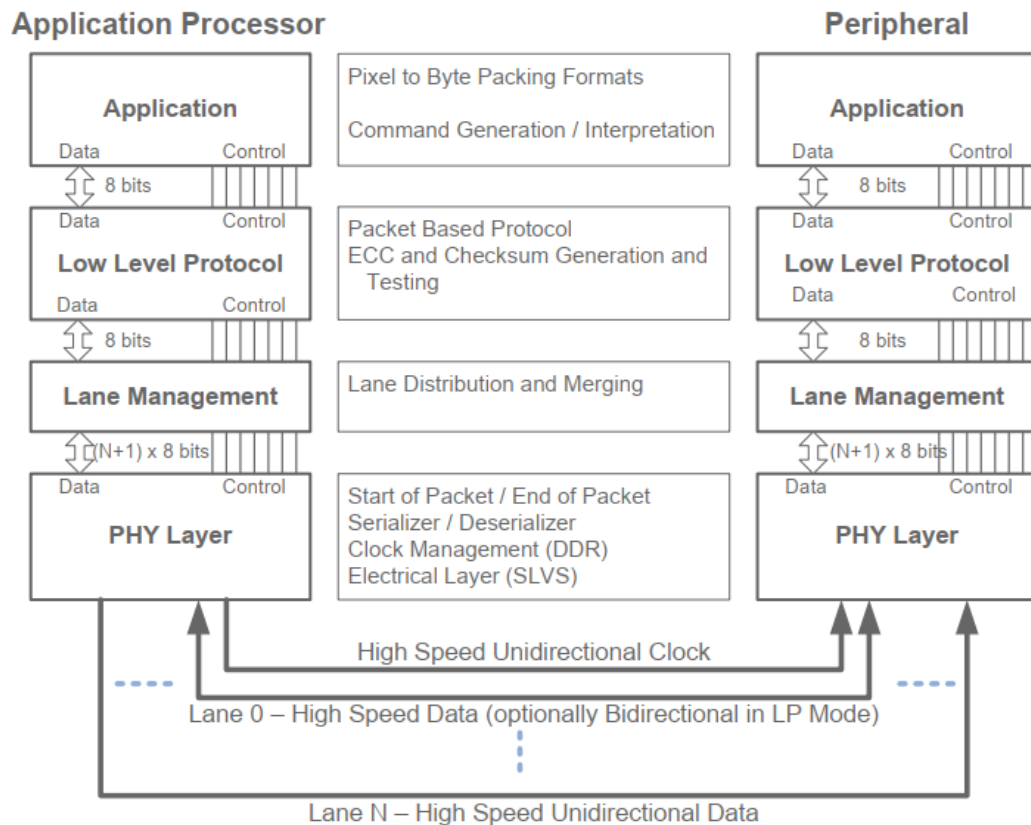
	Timing specification	Absolute time	DA_HS_PREP value (Absolute time/(8*UI))
THS-PREPARE + THS-zero	145ns+10*UI	> 158.33 ns	> 15
THS-PREPARE	40ns+4*UI ~ 85ns +6*UI	45.32 ~ 92.98 ns	5 ~ 8
THS-zero	>60ns+4*UI ~105ns+6*UI	> 112.98~ 65.32 ns	10 ~ 7

- Configure LP Timing Parameter

CONT_DET/LPX/TA_GET/TA_SURE/TA_GO

low power parameters建议使用提供的默认值

DCS(Display Command Set)协议：DCS是一个标准化的命令集，用于命令模式的显示模组。



DSI (Display Serial Interface) 分四层，对应D-PHY、DSI、DCS规范、分层结构图如上：

- PHY 定义了传输媒介，输入/输出电路和和时钟和信号机制。
- Lane Management层：发送和收集数据流到每条lane。
- Low Level Protocol层：定义了如何组帧和解析以及错误检测等。
- Application层：描述高层编码和解析数据流。

3.3. 硬件相关

3.3.1. 查看原理图

Demo Board 原理图:

MIPI Pin Swap

TTL15(MIPI_TX_D3N)	74	PAD_TTL15
TTL14(MIPI_TX_D3P)	73	PAD_TTL14
TTL13(MIPI_TX_D2N)	72	PAD_TTL13
TTL12(MIPI_TX_D2P)	71	PAD_TTL12
TTL11(MIPI_TX_CKN)	70	PAD_TTL11
TTL10(MIPI_TX_CKP)	69	PAD_TTL10
TTL9(MIPI_TX_D1N)	68	PAD_TTL9
TTL8(MIPI_TX_D1P)	67	PAD_TTL8
TTL7(MIPI_TX_D0N)	66	PAD_TTL7
TTL6(MIPI_TX_D0P)	65	PAD_TTL6

SW Development
24310

如上图，芯片内部默认情况下对应的pin是：

Lane0对应的是PAD_TTL6/7 → E_MI_PNL_CH_SWAP_0（对应枚举值是0）

Lane1对应的是PAD_TTL8/9 → E_MI_PNL_CH_SWAP_1（对应枚举值是1）

Lane2对应的是PAD_TTL12/13 → E_MI_PNL_CH_SWAP_3（对应枚举值是3）

Lane3对应的是PAD_TTL14/15 → E_MI_PNL_CH_SWAP_4（对应枚举值是4）

Clk对应的是PAD_TTL10/11 → E_MI_PNL_CH_SWAP_2（对应枚举值是2）

这里的枚举值是软件定义，参考接下来介绍的MIPI Swap栏位配置

3.3.2. 对应padmux

确认硬件设计pin脚连接

如下图是推荐使用的mipi pin，PAD TTL6 – PAD TTL15 (MIPI Mode 1)

SSD201 HW Checklist.xlsx - Excel

文件 开始 插入 页面布局 公式 数据 审阅 视图 福昕PDF xlnhu

剪贴板 字体 对齐方式 数字 样式 单元格 编辑

E1 : default value

	A	B	C	D	J	K	L	M
	Pin Location	Power Domain default 3.3V	Ball Pin Name	GPIO index	reg_tx_mipi_mode	reg_tx_mipi_mode	reg_eth1_mode	reg_eth1_mode
1					1	2	3	5
2					10pin	7pin	9pin	9pin
3								
59	VDDP_1	PAD_TTL3	22					
60	VDDP_1	PAD_TTL4	23					
61	VDDP_1	PAD_TTL5	24					
65	VDDP_1	PAD_TTL6	25	MIPI_TX_P_CH0	MIPI_TX_P_CH0			
66	VDDP_1	PAD_TTL7	26	MIPI_TX_N_CH0	MIPI_TX_N_CH0			
67	VDDP_1	PAD_TTL8	27	MIPI_TX_P_CH1	MIPI_TX_P_CH1			
68	VDDP_1	PAD_TTL9	28	MIPI_TX_N_CH1	MIPI_TX_N_CH1			
69	VDDP_1	PAD_TTL10	29	MIPI_TX_P_CH2	MIPI_TX_P_CH2			
70	VDDP_1	PAD_TTL11	30	MIPI_TX_N_CH2	MIPI_TX_N_CH2			
71	VDDP_1	PAD_TTL12	31	MIPI_TX_P_CH3				
72	VDDP_1	PAD_TTL13	32	MIPI_TX_N_CH3				
73	VDDP_1	PAD_TTL14	33	MIPI_TX_P_CH4				
74	VDDP_1	PAD_TTL15	34	MIPI_TX_N_CH4				
79	VDDP_1	PAD_TTL16	35			ETH1_MDIO		
80	VDDP_1	PAD_TTL17	36			ETH1_MDC		

3.4. 软件相关

3.4.1. 配置padmux

1. user space配置找到项目对应使用的*padmux.dts配置对应MIPI MODE

```
rch/arm/boot/dts/infinity2m-ssc011a-s01a-padmux-display_for_mipi.dtsi - Notepad++
Encoding Language Settings Tools Macro Run Plugins Window ?
File Edit Makefile Makefile infinity2m-ssc011a-s01a-padmux-display_for_mipi.dtsi
<PAD_GPIO11 PINMUX_FOR_GPIO_MODE MDRV_PUSE_I2CSW_SCL>,
<PAD_GPIO12 PINMUX_FOR_GPIO_MODE MDRV_PUSE_I2CSW_SDA>,
<PAD_GPIO13 PINMUX_FOR_GPIO_MODE MDRV_PUSE_I2C1_DEV_IRQ >,
//<PAD_GPIO14 >,
//<PAD_FUART_RX >,
//<PAD_FUART_TX PINMUX_FOR_GPIO_MODE MDRV_PUSE_UTMI_POWER>,
<PAD_FUART_CTS PINMUX_FOR_GPIO_MODE MDRV_PUSE_CPUFREQ_VID0>,
<PAD_FUART_RTS PINMUX_FOR_GPIO_MODE MDRV_PUSE_CPUFREQ_VID1>,
//<PAD_TTL0 PINMUX_FOR_SDIO_MODE_1 MDRV_PUSE_SDIO0_PWR >,

<PAD_TTL6 PINMUX_FOR_TX_MIPI_MODE_1 MDRV_PUSE_TX_MIPI_P_CH0 >,
<PAD_TTL7 PINMUX_FOR_TX_MIPI_MODE_1 MDRV_PUSE_TX_MIPI_N_CH0>,
<PAD_TTL8 PINMUX_FOR_TX_MIPI_MODE_1 MDRV_PUSE_TX_MIPI_P_CH1>,
<PAD_TTL9 PINMUX_FOR_TX_MIPI_MODE_1 MDRV_PUSE_TX_MIPI_N_CH1 >,
<PAD_TTL10 PINMUX_FOR_TX_MIPI_MODE_1 MDRV_PUSE_TX_MIPI_P_CH2 >,
<PAD_TTL11 PINMUX_FOR_TX_MIPI_MODE_1 MDRV_PUSE_TX_MIPI_N_CH2 >,
<PAD_TTL12 PINMUX_FOR_TX_MIPI_MODE_1 MDRV_PUSE_TX_MIPI_P_CH3 >,
<PAD_TTL13 PINMUX_FOR_TX_MIPI_MODE_1 MDRV_PUSE_TX_MIPI_N_CH3 >,
<PAD_TTL14 PINMUX_FOR_TX_MIPI_MODE_1 MDRV_PUSE_TX_MIPI_P_CH4 >,
<PAD_TTL15 PINMUX_FOR_TX_MIPI_MODE_1 MDRV_PUSE_TX_MIPI_N_CH4 >,

<PAD_UART0_RX PINMUX_FOR_UART0_MODE_1 MDRV_PUSE_UART0_RX >,
<PAD_UART0_TX PINMUX_FOR_UART0_MODE_1 MDRV_PUSE_UART0_TX >,
//<PAD_UART1_RX >,
//<PAD_UART1_TX >,
//<PAD_SD_CLK PINMUX_FOR_I2S_MODE_3 MDRV_PUSE_I2S_BCK >,
<PAD_SD_CLK PINMUX_FOR_SDIO_MODE_1 MDRV_PUSE_SDIO0_CLK >,
//<PAD_SD_CMD PINMUX_FOR_I2S_MODE_3 MDRV_PUSE_I2S_SDI >,
<PAD_SD_CMD PINMUX_FOR_SDIO_MODE_1 MDRV_PUSE_SDIO0_CMD >.
```

2. Uboot下padmux配置

MIPI常用的是2Lane跟4Lane，在Uboot下已经根据屏参的Lane num自动切到对应Mode，用户可以不用关注，直接使用即可

```
rsight 4.0 - [hal_pnl.c (Y:\alkaid_mp\boot\drivers\mstar\panel\hal\infinity2m\src)]
Project Options Tools View Window Help
Project (Alt+Shift+P) Advanced... Match Case Word Variations Word Fragments
\boot\include env_common.c (Y:\alkaid_mp\boot\common) env_common.c (Y:\alkaid_stable\boot\common) env_nand.c (Y:\alkaid_mp\boot\common)
322:
323: |
324: void HalPnlSetMipiDisPadMux(HalPnlMipiDsiLaneMode_e enLaneNum)
325: {
326:     W2BYTESMSK(REG_CHIPTOP_09_L, 0x0000, 0xFFFF);
327:
328:     W2BYTESMSK(REG_CHIPTOP_0F_L, 0x0000, 0xFFFF);
329:     if (enLaneNum == E_HAL_PNL_MIPI_DSI_LANE_2)
330:     {
331:         W2BYTESMSK(REG_CHIPTOP_0D_L, 0x2000, 0xFFFF); // MipiDsi 2Lane
332:     }
333:     else
334:     {
335:         W2BYTESMSK(REG_CHIPTOP_0D_L, 0x1000, 0xFFFF); // MipiDsi 4Lane
336:     }
337: }
338:
339:
340:
```

3.4.2. 配置屏参

确认panel规格：

6.2 Timing Conditions

Item	Symbol	Values			Unit	Remark
		Min	Typ	Max		
Horizontal Display Area	thd	-	1024	-	DCLK	
DCLK frequency	fck	44.9	51.2	63	MHz	
One horizontal line	th	1200	1344	1400	DCLK	
HS pulse width	thpw	1	-	140	DCLK	
HS Blanking	thb	160	160	160	DCLK	
HS Front Porch	thfp	16	160	216	DCLK	

Item	Symbol	Values			Unit	Remark
		Min	Typ	Max		
Vertical display Area	tvd	-	600	-	TH	
VS period time	tv	624	635	750	TH	
VS pulse width	tvpw	1	-	20	TH	
VS Blanking	tvb	23	23	23	TH	
VS Front Porch	tvfp	1	12	127	TH	

$H_{Total} = u16Width + u16HSyncWidth + u16HSyncBackPorch + HSyncFrontPorch$

$V_{Total} = u16Height + u16VSyncWidth + u16VSyncBackPorch + VSyncFrontPorch$

$DCLK = H_{Total} * V_{Total} * FPS$

3.4.3. 代码示例

MIPI屏参在配置上会比TTL的稍有差异，主要关注两个结构体，注意参数设置：

- 1. 配置MI_PANEL_MipiDsiConfig_t (TTL不需要关注)

```

1. typedef struct
2. {
3.     MI_U8 u8HsTrail;
4.     MI_U8 u8HsPrpr;
5.     MI_U8 u8HsZero;
6.     MI_U8 u8ClkHsPrpr;
7.     MI_U8 u8ClkHsExit;
8.     MI_U8 u8ClkTrail;
9.     MI_U8 u8ClkZero;
10.    MI_U8 u8ClkHsPost;
11.    MI_U8 u8DaHsExit;
12.    MI_U8 u8ContDet;
13.
14.    MI_U8 u8Lpx;
15.    MI_U8 u8TaGet;
16.    MI_U8 u8TaSure;
17.    MI_U8 u8TaGo;
18.
19.    MI_U16 u16Hactive;
20.    MI_U16 u16Hpw;
21.    MI_U16 u16Hbp;
22.    MI_U16 u16Hfp;
23.
24.    MI_U16 u16Vactive;
25.    MI_U16 u16Vpw;
26.    MI_U16 u16Vbp;
27.    MI_U16 u16Vfp;
28.
29.    MI_U16 u16B11p;
30.    MI_U16 u16Fps;
31.
32.    MI_PANEL_MipiDsiLaneMode_e enLaneNum;
33.    MI_PANEL_MipiDsiFormat_e enFormat;
34.    MI_PANEL_MipiDsiCtrlMode_e enCtrl;
35.
36.    MI_U8 *pu8CmdBuf;
37.    MI_U32 u32CmdBufSize;
38.
39.    MI_U8 u8SyncCalibrate;
40.    MI_U16 u16VirHsyncSt;
41.    MI_U16 u16VirHsyncEnd;
42.    MI_U16 u16VsyncRef;
43.    MI_U16 u16DataClkSkew;
44.
45.    MI_U8 u8PolCh0; // channel 0 polarity, 0:HW default, 1:positive, 2:negative
46.    MI_U8 u8PolCh1; // channel 1 polarity, 0:HW default, 1:positive, 2:negative
47.    MI_U8 u8PolCh2; // channel 2 polarity, 0:HW default, 1:positive, 2:negative
48.    MI_U8 u8PolCh3; // channel 3 polarity, 0:HW default, 1:positive, 2:negative
49.    MI_U8 u8PolCh4; // channel 4 polarity, 0:HW default, 1:positive, 2:negative
50.
51.    MI_PANEL_MipiDsiPacketType_e ePacketType;
52.}MI_PANEL_MipiDsiConfig_t;

```

Part1

Part2

Part3

Part4

- Part2主要是设置Panel规格书列明的H/V相关参数，参照TTL的方法逐个填入接可
- Part3主要是Panel的初始化cmd，这部份cmd是各panel厂商参照DSC的标准自定义的，所以cmd一般是找Panel的芯片原厂提供

```

1. static MI_U8 cmd_buf[] =
2. {
3.     cmd1, data_num, data[0],data[1],...,data[num-1],
4.     ...,
5.     cmdN, data_num, data[0],data[1],...,data[num-1],
6.     FLAG_END_OF_TABLE, FLAG_END_OF_TABLE, //结束符
7. }

```

- Part4是设置各Lane的P/N极性翻转，参考硬件跟Panel的连接确认是否需要翻转即可
- Part1这部分主要是对传输过程中时序要求，这部分我们准备了一个通用的Excel表格(附件：[mipi相关参数计算表格v2.xlsx](#) [media/mipi相关参数计算表格v2.xlsx])，只要输入对应Panel的HTotal/VTotat即会自动生成，示例如下：

红框部分就是按照公式自动生成的惨呼，在范围内微调对应填到软件屏参即可

Parameter	Absolute time		stMipiDsiConfig	
	min(ns)	max(ns)	min	max
HsTrail	max(8n*UI, 60ns+4n*UI)	69.92	4	
HsPrpr	40ns+4UI~85ns+6UI	49.92 99.88	3	5
HsZero	60ns+4UI~105ns+6UI	69.92 119.88	4	6
HsPrpr+HsZero	145ns+10UI~	169.8	9	
ClkHsPrpr	38ns~95ns	38 95	2	4
ClkHsExit	100ns~	100	6	
ClkTrail	60ns~	60	4	
ClkZero	300ns-ClkHsPrpr	205 262	11	13
ClkHsPost	60ns+52UI~	188.96	10	
DaHsExit	100ns~	100	6	

2. MI_PANEL_ParamConfig_t：主要看MIPI Swap配置栏位 (TTL不需要关注)

```

1. typedef enum
2. {
3.     E_MI_PNL_CH_SWAP_0,
4.     E_MI_PNL_CH_SWAP_1,
5.     E_MI_PNL_CH_SWAP_2,
6.     E_MI_PNL_CH_SWAP_3,
7.     E_MI_PNL_CH_SWAP_4,
8. }MI_PANEL_ChannelSwapType_e;
9.
10. typedef struct
11. {
12.     ...
13.     ...
14.     MI_PANEL_ChannelSwapType_e eCh0;    //--> CLK 栏位

```

```

15.     MI_PANEL_ChannelSwapType_e eCh1;    //--> Lane3栏位
16.     MI_PANEL_ChannelSwapType_e eCh2;    //--> Lane2栏位
17.     MI_PANEL_ChannelSwapType_e eCh3;    //--> Lane1栏位
18.     MI_PANEL_ChannelSwapType_e eCh4;    //--> Lane0栏位
19. }MI_PANEL_ParamConfig_t;

```

需要参考实际的硬件连接配置，考虑外接设备Lane To Lane不做Swap的话，参数应配置如下：

```

1.  typedef struct
2.  {
3.      ...
4.      ...
5.      MI_PANEL_ChannelSwapType_e eCh0;    //--> CLK  栏位
-- E_MI_PNL_CH_SWAP_2
6.      MI_PANEL_ChannelSwapType_e eCh1;    //--> Lane3栏位
-- E_MI_PNL_CH_SWAP_4
7.      MI_PANEL_ChannelSwapType_e eCh2;    //--> Lane2栏位
-- E_MI_PNL_CH_SWAP_3
8.      MI_PANEL_ChannelSwapType_e eCh3;    //--> Lane1栏位
-- E_MI_PNL_CH_SWAP_1
9.      MI_PANEL_ChannelSwapType_e eCh4;    //--> Lane0栏位
-- E_MI_PNL_CH_SWAP_0
10. }MI_PANEL_ParamConfig_t;

```

即2, 4, 3, 1, 0

屏参示例：

```

MI_PANEL_MipiDsiConfig_t stMipiDsiConfig =
{
    //HsTrail HsPrpr HsZero ClkHsPrpr ClkHsExit ClkTrail ClkZero ClkHsPost DaHsExit ContDet
    0x05, 0x03, 0x05, 0x0A, 0x0E, 0x03, 0x0B, 0x0A, 0x05, 0x00,
    //Lpx  TaGet  TaSure  TaGo
    0x10, 0x1a, 0x16, 0x32,

    //Hac,  Hpwr,  Hbp,  Hfp,  Vac,  Vpw,  Vbp,  Vfp,  Bllp, Fps
    800,  20,  80,  60,  1280,  10,  20,  20,  0,  60,

    E_MI_PNL_MIPI_DSI_LANE_4,    // MIPnlMipiDsiLaneMode_e enLaneNum;
    E_MI_PNL_MIPI_DSI_RGB888,    // MIPnlMipiDsiFormat_e enFormat;
    E_MI_PNL_MIPI_DSI_SYNC_PULSE, // MIPnlMipiDsiCtrlMode_e enCtrl;

    xxx_CMD,
    sizeof(xxx5_CMD),
    1, 0x01AF, 0x01B9, 0x80D2, 7,|
    .ePacketType = E_MI_PNL_MIPI_DSI_PACKET_TYPE_DCS,
};

```

```

35. int sstar_disp_init(MI_DISP_PubAttr_t *pstDispPubAttr, int
disp_port_num)
36. {
37.     MI_PANEL_LinkType_e eLinkType;

```

```

38.
39.     if (pstDispPubAttr->eIntfType == E_MI_DISP_INTF_LCD)
40.     {
41.         pstDispPubAttr->stSyncInfo.u16Vact =
stPanelParam.u16Height;
42.         pstDispPubAttr->stSyncInfo.u16Vbb =
stPanelParam.u16VSyncBackPorch;
43.         pstDispPubAttr->stSyncInfo.u16Vfb =
stPanelParam.u16VTotal - (stPanelParam.u16VSyncWidth +
44.
stPanelParam.u16Height + stPanelParam.u16VSyncBackPorch);
45.         pstDispPubAttr->stSyncInfo.u16Hact =
stPanelParam.u16Width;
46.         pstDispPubAttr->stSyncInfo.u16Hbb =
stPanelParam.u16HSyncBackPorch;
47.         pstDispPubAttr->stSyncInfo.u16Hfb =
stPanelParam.u16HTotal - (stPanelParam.u16HSyncWidth +
48.
stPanelParam.u16Width + stPanelParam.u16HSyncBackPorch);
49.         pstDispPubAttr->stSyncInfo.u16Bvact = 0;
50.         pstDispPubAttr->stSyncInfo.u16Bvbb = 0;
51.         pstDispPubAttr->stSyncInfo.u16Bvfb = 0;
52.         pstDispPubAttr->stSyncInfo.u16Hpw =
stPanelParam.u16HSyncWidth;
53.         pstDispPubAttr->stSyncInfo.u16Vpw =
stPanelParam.u16VSyncWidth;
54.         pstDispPubAttr->stSyncInfo.u32FrameRate =
stPanelParam.u16DCLK * 1000000 / (stPanelParam.u16HTotal *
stPanelParam.u16VTotal);
55.         pstDispPubAttr->eIntfSync = E_MI_DISP_OUTPUT_USER;
56.         pstDispPubAttr->eIntfType = E_MI_DISP_INTF_LCD;
57.         eLinkType = stPanelParam.eLinkType;
58.
59.         MI_DISP_SetPubAttr(0, pstDispPubAttr);
60.         MI_DISP_Enable(0);
61.
62.         MI_PANEL_Init(eLinkType);
63.         MI_PANEL_SetPanelParam(&stPanelParam);
64.         if(eLinkType == E_MI_PNL_LINK_MIPI_DSI)
65.         {
66.             MI_PANEL_SetMipiDsiConfig(&stMipiDsiConfig);
67.         }
68.
69.     }
70.     return 0;
71. }

```

3.5. Debug方法

cat /proc/mi_modules/mi_panel/mi_panel0

```
/ # cat /proc/mi_modules/mi_panel/mi_panel0
PanelName:SAT070CP50_800x480_60

----- PANEL Dev0 Info -----
LVDS_POL      LVDS_CH      LINK_TYPE      TI_MODE
    0           0          TTL          1
SW_ODD        SW_EVEN      SW_ODD_RB      SW_EVEN_RB
    0           0           0           0
H_Total      V_Total      Width      Height      H_Start      V_Start
  1344        635      1024      600         98         27
  hbp        hspw        hfp        vbp         vspw         vfp
    46         48        226        23          4          8
  DClk      FrameRate      INV_DCLK      INV_DE      InvHSync      InvVSync
    51          0           0           0          0          0
SSC_Enable   SSC_Step      SSC_Span      TI_BIT      Format      chnswap
enable       25         192        8BIT      8BIT      (0,1,2,3,4)
```

4. HDMI输出介绍

4.1. 规格

HDMI，英文全称是High DefinitionMultimedia Interface，中文名称是高清晰多媒体接口的缩写

HDMI 1.4b transmitter up to 4k2k@30

常规分辨率：

- DACOUT_480P_60
- DACOUT_576P_50
- DACOUT_720P_50
- DACOUT_720P_60
- DACOUT_1024X768P_60
- DACOUT_1080P_24
- DACOUT_1080P_25

DACOUT_1080P_30

DACOUT_1080P_50

DACOUT_1080P_60

DACOUT_1920X1200P_60

DACOUT_1280X1024P_60

DACOUT_1280X800P_60

DACOUT_1366X768P_60

DACOUT_1440X900P_60

DACOUT_1600X1200P_60

DACOUT_1680X1050P_60

DACOUT_4K2KP_30

4.2. 基本概念

- 标准HDMI连接器

Pin	Signal
1	TMDS Data2+
2	TMDS Data2 Shield
3	TMDS Data2
4	TMDS Data1+
5	TMDS Data1 Shield
6	TMDS Data1–
7	TMDS Data0+
8	TMDS Data0 Shield
9	TMDS Data0–
10	TMDS Clock+
11	TMDS Clock Shield
12	TMDS Clock–
13	CEC
14	Reserved (N.C. on device)
15	SCL
16	SDA
17	DDC/CEC Ground
18	+ 5V
19	Hot Plug Detect

- DDC通道

DDC全文为Display Data Channel

传送端与接收端可利用DDC通道得知彼此的传送与接收能力，但HDMI仅需单向获知接收端(显示器)的能力

使用100kHz时钟频率的I²C信号

传送数据结构为VESA Enhanced EDID (V1.3)

- TMDS通道

传送音频，视频，以及各种辅助数据

信号编码方式：遵循DVI 1.0规格。Single-link (Type A HDMI)

视频像素带宽：每个像素的容许数据量从24位至48位。支持每秒60张画面1080p

像素编码方式：RGB 4:4:4, YCbCr 4:4:4; YCbCr 4:2:2

音频采样率：32 kHz, 48 kHz

音频声道数量：最大2声道

音频流规格：PCM Audio/16BIT

- HPD

HPD (Hotplug热插拔) 是由接收端设备 (Sink) 发出的，在与HDMI源端设备 (Source) 之间建立正式通信的前奏信号。HPD信号电平为高时表示接收端设备已经准备好了，允许源端设备访问接收端设备。

4.3. 点屏流程

HDMI接口为固定接口，请参考如下硬件和软件配置。

4.4. 硬件相关

IC pin接口：

4.5.2. 初始化和参数设定

Hdmi相关代码：

```
/* Base Module Init */
stInitParam.pCallbackArgs = NULL;
stInitParam.pfnHdmiEventCallback = Hdmi_callback_impl;
ExecFunc(MI_HDMI_Init(&stInitParam), MI_SUCCESS);
ExecFunc(MI_HDMI_Open(eHdmi), MI_SUCCESS);

ExecFunc(MI_HDMI_SetAttr(eHdmi, &stAttr), MI_SUCCESS);
ExecFunc(MI_HDMI_Start(eHdmi), MI_SUCCESS);
//ExecFunc(MI_HDMI_SetAvMute(eHdmi, true), MI_SUCCESS); // first no
audio
ExecFunc(MI_HDMI_SetAvMute(eHdmi, false), MI_SUCCESS);
```

Display与hdmi接口：

```
memset(&stPubAttr, 0x00, sizeof(MI_DISP_PubAttr_t));
stPubAttr.u32BgColor = YUYV_BLACK;
stPubAttr.eIntfType = E_MI_DISP_INTF_HDMI;
stPubAttr.eIntfSync = eOutputTiming;
/* Set Disp Pub */
ExecFunc(MI_DISP_SetPubAttr(DispDev, &stPubAttr), MI_SUCCESS);
ExecFunc(MI_DISP_Enable(DispDev), MI_SUCCESS);
```

Hdmi参数：

```
MI_HDMI_Attr_t stAttr;

stAttr.stEnInfoFrame.bEnableAudInfoFrame = true;
stAttr.stEnInfoFrame.bEnableAviInfoFrame = true;
stAttr.stEnInfoFrame.bEnableSpdInfoFrame = true;
stAttr.stAudioAttr.bEnableAudio = true;
stAttr.stAudioAttr.bIsMultiChannel = 0;
stAttr.stAudioAttr.eBitDepth = E_MI_HDMI_BIT_DEPTH_16;
stAttr.stAudioAttr.eCodeType = E_MI_HDMI_ACODE_PCM;
stAttr.stAudioAttr.eSampleRate = E_MI_HDMI_AUDIO_SAMPLERATE_48K;
stAttr.stVideoAttr.bEnableVideo = true;
stAttr.stVideoAttr.eColorType =
E_MI_HDMI_COLOR_TYPE_RGB444; //default color type
stAttr.stVideoAttr.eDeepColorMode = E_MI_HDMI_DEEP_COLOR_MAX;
stAttr.stVideoAttr.eOutputMode = E_MI_HDMI_OUTPUT_MODE_HDMI;
stAttr.stVideoAttr.eTimingType =
```

```
E_MI_HDMI_TIMING_1080_60P;//1080P60 -> 720P60  
s32ret = MI_HDMI_SetAttr(eHdmi, &stAttr);
```

5. SPI点屏介绍

5.1. 规格

规格 芯片	SPI
201/202	支持
203	支持

5.2. 基本概念

SPI：Serial Peripheral Interface，串行外设接口。

利用spi接口一般点比较小分辨率的lcd。

5.3. 点屏流程

1. 根据原理图配置SPI PIN脚。查询CheckList表格并更改dts文件，配置对应管脚的PIN_MUX。
2. Kernel中使能SPI驱动

menuconfig下使能如下选项：

```
> Device Drivers  
  > Sstar Soc platform drivers  
    > Sstar MSPI driver
```

- 3. 根据数据手册下初始化命令。初始化命令一般由屏厂提供，或者查看屏规格书自行配置。
- 4. 写入图像数据，显示画面。有些屏自带内部测试partten，可通过命令使能输出。

5.4. 硬件相关

5.4.1. 查看原理图

以SPI为例，硬件工程师在绘制原理图时会查询HW CheckList表格，选择一组空闲的，并且可以作为SPI接口的管脚。我们选择PAD_GPIO8-PAD_GPIO11，并且PADMUX选择SPI0_MODE_5。

5.4.2 对应padmux

Ball Pin Name	GPIO index	reg_spi0_mode
		5
		4pin
PAD_GPIO5 (PWM1)	5	
PAD_GPIO6 (I2C0_SCL)	6	
PAD_GPIO7 (I2C0_SDA)	7	
PAD_GPIO8 (SPI0_CZ)	8	SPI0_CZ
PAD_GPIO9 (SPI0_CK)	9	SPI0_CK
PAD_GPIO10 (SPI0_DI)	10	SPI0_DI
PAD_GPIO11 (SPI0_DO)	11	SPI0_DO
PAD_GPIO12	12	

5.5. 软件相关

5.5.1. 配置相关padmux

在已知管脚的PADMUX情况下，我们需要更改kernel/arch/arm/boot/dts目录下对应的padmux.dtsi文件，保证PADMUX切换正确。以SPI0_MODE_5为例：

```
<PAD_GPIO8 PINMUX_FOR_SPI0_MODE_5 MDRV_PUSE_SPI0_CZ >,  
<PAD_GPIO9 PINMUX_FOR_SPI0_MODE_5 MDRV_PUSE_SPI0_CK >,  
<PAD_GPIO10 PINMUX_FOR_SPI0_MODE_5 MDRV_PUSE_SPI0_DI >,  
<PAD_GPIO11 PINMUX_FOR_SPI0_MODE_5 MDRV_PUSE_SPI0_DO>,
```

5.5.2. 配置屏参

SPI没有我们TTL类似的屏参配置，主要是根据数据手册下初始化命令。初始化命令一般由屏厂提供，或者查看屏规格书自行配置。

5.5.3. 代码示例

SPI屏有两种开发方式：

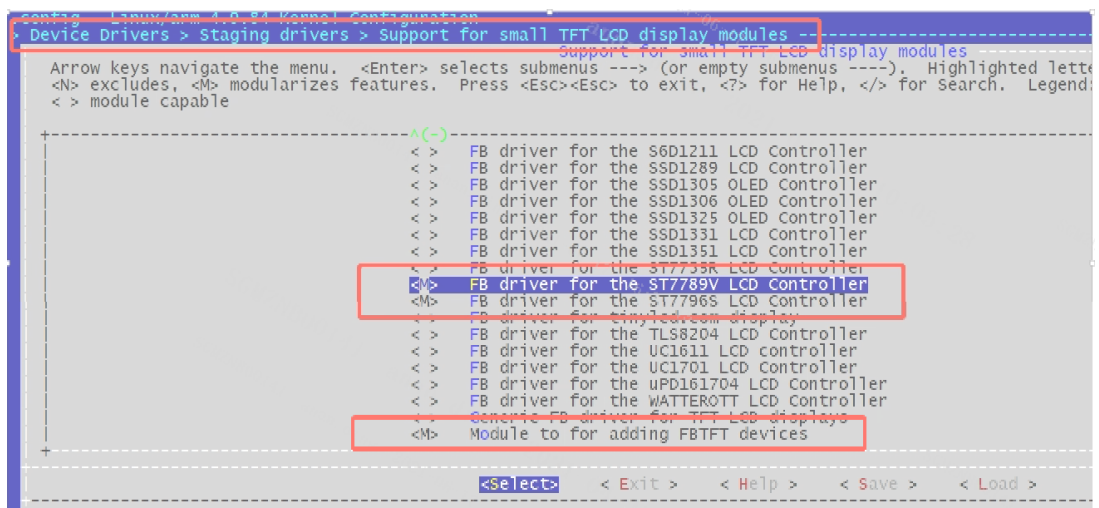
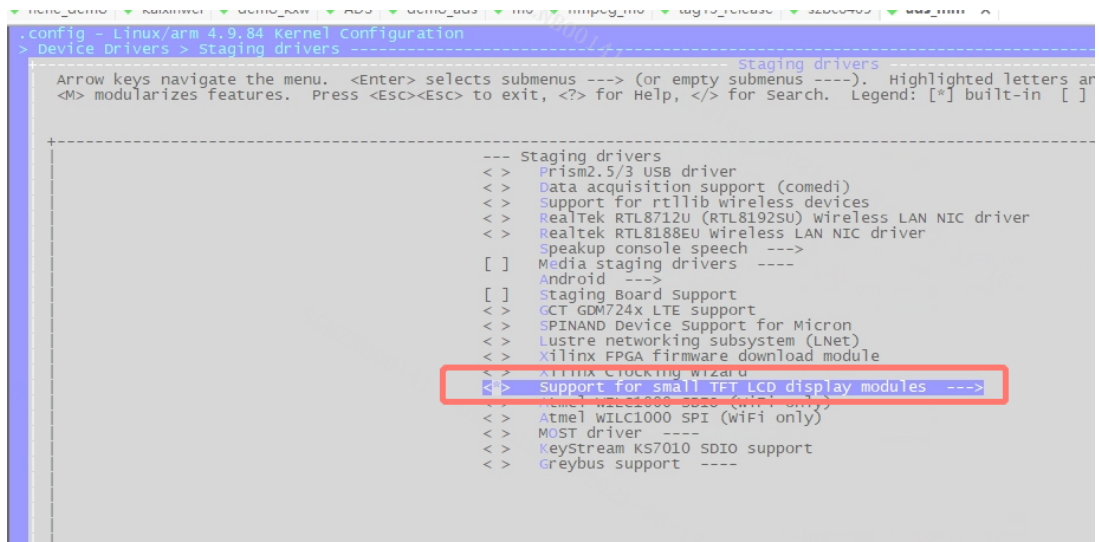
- a. 直接将YUV/RGB数据通过SPI发送到panel端，这个操作参考BSP中SPI使用参考发数据即可(下文不在赘述)
 - b. kernel原生框架有通过fbtft底层对接spi，用户操作只需要操作fbtft抽象出来的/dev/fb1即可，具体使用参考章节5.6。
-

5.6. Fbtft开发建议

对于spi panel建议使用kernel原生的架构去开发，使用TFT-LCD驱动。

5.6.1. Spi panel驱动配置

1. Kernel menuconfig中打开fbtft config，把图中标记编译成模块



注意：Support for small TFT LCD display这个需要设置为 *

2. Dts添加对应配置

```

spi1@1{
    compatible = "ssstar,mspi";
    mspi-group = <1>;
#ifdef CONFIG_CAM_CLK
    camclk = <CAMCLK_mspi>;
#else
    clocks = <&CLK_mspi>;
#endif
    reg = <0x1F222200 0x200>;
    interrupts = <GIC_SPI INT_IRQ_MSPI_1 IRQ_TYPE_LEVEL_HIGH>;
    use-dma = <0>;
    status = "ok";
    admatec_b-berry28@0 {
        compatible = "sitronix,st7789v";
        reg = <0>;
        status = "ok";
        spi-max-frequency = <9000000>;
        rotate = <0>;
        fps = <30>;
        buswidth = <8>;
        reset-gpio = <&gpio 21 0>;
        dc-gpio = <&gpio 20 0>;
        debug = <0>;
    };
};

```

编译后insomod安装，命令如下：

- 移出原有的fbdev.ko，重启平台
- 重启后加载fbtft模块

```

insmod /config/modules/4.9.84/fbtft_device.ko busnum=1
name=admatec_b-berry28 cs=0 gpios="reset:21,dc:20" custom=1
speed=40000000 width=240 height=320 buswidth=8 mode=0
debug=7 dma=0

```

```

00274:
00275: /* Supported displays in alphabetical order */
00276: static struct fbtft_device_display displays[] = {
00277: {
00278:     .name = "adafruit18",
00279:     .spi = &(struct spi_board_info) {
00280:         .modalias = "fb_st7735r",
00281:         .max_speed_hz = 32000000,
00282:         .mode = SPI_MODE_0,
00283:         .platform_data = &(struct fbtft_platform_data) {
00284:             .display = {
00285:                 .buswidth = 8,
00286:                 .backlight = 1,
00287:             },
00288:             .gpios = (const struct fbtft_gpio []) {
00289:                 { "reset", 25 },
00290:                 { "dc", 24 },
00291:                 { "led", 18 },
00292:             },
00293:             .gamma = ADAFRUIT18_GAMMA,
00294:         },
00295:     },
00296: };

```

注意：name=admatec_b-berry28 要和驱动fbtft_device_display displays[] 的name一致。Speed参数如果过高可能会出现spi timeout，可以尝试降低speed做调试。

- 加载对应TFT驱动: insmod /config/modules/fb_st7796s.ko

3. 针对panel的相关设定的更改

a. 屏初始化

Fbtf-core.c:

fbtft_probe_common() -> fbtft_register_framebuffer() ->

fbtftops.init_display()

会使用如下数组的初始化参数对panel做初始化:

```
}, {  
    .name = "admatec_b- berry28",  
    .spi = &(struct spi_board_info) {  
        .modalias = "fb_st7789v",  
        .max_speed_hz = 48000000,  
        .mode = SPI_MODE_0,  
        .platform_data = &(struct fbtft_platform_data) {  
            .display = {  
                .buswidth = 8,  
                .backlight = 1,  
                .init_sequence = cberry28_init_sequence,  
            },  
            .gpios = (const struct fbtft_gpio []) {  
                { "reset", 21 },  
                { "dc", 20 },  
                { "led", 5 },  
            },  
        },  
        .gamma = CBERRY28_GAMMA,  
    },  
}
```

其他设定分别在

fbtftops.set_var()/fbtftops.update_display()fbtftops.set_gamma()

b. 刷新

上层通过ioctl(g_fbFd, FBIOPAN_DISPLAY, &vinfo)来刷新fb

对应底层:

fbtft_update_display() -> fbtftops.write_vmem() ->

fbtft_write_vmem16_bus8() -> fbtftops.write -> fbtft_write_spi()

Note: Because the spi sends data directly to the panel, the data sent by RGB565 or RGB888 is different, so the data data sent by the upper layer to the driver needs to conform to the data format of the screen. If the flythings used by the gui need to close the gfx module: export ZK_GFX_ENABLE=0

5.7. Debug method

If the SPI Panel does not light up, you can check it from the following methods:

- a. Is the padmux of the spi configured correctly? It can be confirmed by looking at registers and measuring waveforms in hardware.
 - b. Whether the spi initialization command is correct, you can contact the screen factory to confirm, and you can send the spi command to play the test pattern to test whether the hardware is connected.
-

6. Frequently Asked Questions

The backlight can be on, but there is no picture, please check the following problems:

- a. Check whether the screen parameters meet the panel spec requirements
- b. Whether the data flow is normal and whether there is input buff input
- c. Whether the relevant settings of DISP device/layer/input port are reasonable
- d. Whether the panel init is successful, the TTL panel that needs SPI initialization can measure whether the SPI signal is normal and whether the initial cmd is correctly transmitted
- e. Check whether the polarity of Hsync/Vsync/DE/Dclk meets the panel spec
- f. MIPI panel check whether the line sequence of clk lane/data lane is consistent with the software setting
- g. Check whether riu_r 0x101e 0xd has the correct TTL mode(value:0x100)/MIPI mode(value:0x1000)